Bing GAO

Computational Mechanics Engineering Student | CFD, Thermal Systems & Engineering Software
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Summary

Engineer with nearly 7 years in mobile software development and technical leadership, now applying 2M+ user product delivery experience to CFD, FEM, thermal modelling, and numerical verification while completing a Diplôme d'Ingénieur.

Experience

Vinci Eco Drive, ESILV Formula Student — Paris La Défense, France
Aerodynamics Engineer — 09/2025 - Present

- Led thermal-system modelling for the battery, inverter, and motor, coupling hydraulic, radiator, and transient coolant models to screen the E3 cooling architecture before procurement.
- Developed cooling-duct concepts within the aerodynamic package and used an OpenFOAM fan/core surrogate to qualify the numerical method before detailed vehicle CFD.

Felisiak Ingénierie & Développement — Paris, France
3D Modelling Intern — 04/2026 - 08/2026

- Reconstructed the ESA Space Rider from public blueprints for a launch-campaign review application and delivered separate review and animation-ready assets for Unity and Blender.
- Built a scripted Blender audit, rollback, and export workflow and prepared the engineering report, defense poster, and browser-ready GLB.

Inkeverse Group Ltd — Beijing, China
Mobile Application Development Manager — 08/2016 - 03/2023

- **Product Leadership:** Led Lao You from inception to 2M+ users; built the product from scratch, wrote 70% of its initial core code, and helped establish it as a revenue source.
- **Algorithms & Audio:** Independently developed voiceprint-recognition algorithms and real-time gift-rendering effects for live streaming.
- **DevOps Automation:** Designed a Ruby-based CI/CD pipeline that automated testing, packaging, and distribution across Android and iOS.
- **Security Architecture:** Implemented code/resource obfuscation and SSL pinning to increase resistance to reverse engineering and traffic interception.
- **Hybrid Stack:** Integrated React Native and built a hotfix mechanism for controlled runtime updates.

Education

ESILV — Diplôme d'Ingénieur (expected 2027)
Modelling & Computational Mechanics — Paris La Défense, France • 09/2025 - 2027

- Coursework: Computational Fluid Dynamics and finite-volume methods, Finite Element Method, Vehicle Dynamics and Powertrains, Sustainable Automotive Engineering, Structural Optimization, Machine Learning and Digital Twins.

Alliance Française de Beijing — Full-time French Studies
Beijing, China • 04/2023 - 05/2024

Changchun University of Science and Technology — Bachelor • 3.33/4.00
Computer Science — Changchun, China • 09/2012 - 07/2016

- **Patent:** "Mouse with gamepad function" (CN203870574U, issued Oct 8, 2014).
- **1st Provincial Prize:** China Undergraduate Mathematical Contest in Modeling (2014).
- **1st Prize:** 8th CUST ACM Programming Competition (2013).
- **National Scholarship:** Ministry of Education award, top 0.2% nationwide (2014-2015).
- **First Class Scholarship:** Awarded 5 times for academic excellence (2012-2015).

Skills

Simulation & Numerical Methods
OpenFOAM, Ansys Fluent, Ansys Mechanical, Star-CCM+, Abaqus, CFD (FVM/RANS), FEM, Lattice Boltzmann Method, POD/DMD, thermal-fluid modelling, mesh and convergence studies

CAD & 3D
CATIA, SolidWorks, Blender

Programming
Java/Kotlin, Python, Ruby/Ruby on Rails, C++, C, Rust, TypeScript/React, Node.js, MATLAB/Simulink

Languages
French (DELF B2), English (C1), Chinese (native)

Community

Google Developer Group Beijing — 11/2015 - 03/2023
Beijing, China
Core Organizer (part-time)

- **Community Leadership:** Organized free technical events that regularly drew 200+ attendees and featured speakers from Google and Silicon Valley.
- **Developer Advocacy:** Led technical knowledge sharing across mobile development, open-source tools, and emerging engineering topics.

Selected Engineering Projects

Formula Student Cooling System & OpenFOAM Surrogate Vinci Eco Drive | 2025-2026

- Coupled vendor fan, radiator, and pump curves with an 80-cell one-dimensional finite-volume coolant model; the hydraulic operating point reached 10.03 L/min.
- Identified that the passive E3 architecture violated the inverter temperature boundary before procurement and detailed vehicle CFD, documenting a NO-GO decision and replacement architecture paths.
- Qualified a fan-jump and porous-core OpenFOAM surrogate on 4 meshes from 7,440 to 3,481,920 cells; fine-to-extra-fine changes were 0.950% in flow and 0.0145% in pressure loss.
- Scope: numerical screening and surrogate-method qualification, not installed-vehicle airflow, heat-transfer, hardware, or track validation.

F1 External-Aerodynamics RANS Pilot Independent study | 2026

- Built a 4.35M-cell OpenFOAM 14 half-car RANS baseline and completed a 23-variant pilot campaign across geometry, boundary-condition, and modelling sensitivities.
- Automated CAD variant staging, snappyHexMesh, steady RANS, validity gates, component-force extraction, and result archiving; retained 9 valid sensitivity cases and preserved 2 diverged roughness runs.
- Kept material mesh sensitivity visible and excluded absolute aerodynamic-performance, yaw, wind-tunnel, and physical-correlation claims.

FlowLab & FlowROM Independent study | 2026

- Implemented a dependency-free D2Q9 BGK lattice-Boltzmann solver shared by Node.js and the browser; validated 3 grids at Re=100 against canonical lid-driven-cavity data.
- Generated 480 velocity fields, using 320 snapshots for training and 160 snapshots across 4 complete holdout cycles for POD/DMD evaluation.
- Achieved 0.123% rank-8 POD holdout error with 48.8x compression and 0.100% DMD full-state holdout error; scope remains controlled two-dimensional internal flow.

Space Rider Digital Model Felisiak Ingénierie & Développement | 2026

- Reconstructed the ESA Space Rider from public blueprints during a 17-week internship; the internal verification record reports +/-6 mm side-view and +/-8.4 mm top-view agreement over a 4.88 m body.
- Used scripted geometry audits, protected baselines, rollback branches, and reference-view checks to retain failed approaches and prevent local surface fixes from changing accepted proportions.
- Delivered separate review and animation-ready assets and reduced the browser GLB from 11.6 MB to 639 KB while preserving the declared release structure.
- Scope: reference-led work for Felisiak using public ESA material, not ESA employment, manufacturer CAD, or flight-article validation.

XC48 Abaqus Tensile Twin Course project | 2026

- Built a Python pipeline from raw force/displacement data to validated material inputs, Abaqus ODB post-processing, and report generation for an XC48 tensile study.
- Matched the retained experiment with R^2=0.9663 on the M2 mesh and verified a 5% pre-fracture kinetic/internal energy bound; scope excludes sole authorship of the full team Abaqus model.

Airfoil & Ground-Effect Methods Independent studies | 2026

- Implemented a cosine-spaced Hess-Smith panel method for NACA 0012 and compared linear lift against NASA TM-4074; the 160-to-240-panel lift change at 4 degrees was 0.0307%.
- Built a horseshoe/image-vortex ground-effect solver over 14 h/c cases; recovered free-air lift within 0.0034% at h/c=50 and changed 0.258% from 64 to 96 span panels.
- Scope: linear, inviscid, steady methods, without viscous drag, separation, stall, tyres, bodywork, or race-car performance claims.